

toxicity was pulmonary, occurring in 2-7% in most series, but in as many as 13-18% of patients in other series. Pulmonary toxicity occurred as alveolitis or fibrosis and has been fatal in about 10% of reported patients. This overall rate may overstate the true fatality rate in a carefully monitored patient population because some of the deaths occurred before the pulmonary toxicity of amiodarone was recognized as a drug effect and therefore patients were continued on treatment despite toxicity. Nonetheless, even carefully monitored patients can die of pulmonary toxicity. Recent studies have shown frequencies of overt pulmonary toxicity of 10% and 15%, and some degree of diffusion defect appears in more than 50% of patients.

Although most patients in the investigational studies developed visible accumulations of amiodarone in the cornea, impaired vision was relatively uncommon, occurring in up to 10% of patients (mostly as the appearance of halos around objects). Mild liver dysfunction was common, occurring in 5-40% in some studies, but it was rarely a cause of discontinuation. However, fatal liver injury has been reported. Photosensitivity of the skin and blue discoloration were also common.

Amiodarone, like other antiarrhythmics, can worsen arrhythmias. Worsened arrhythmias were seen in 2-9% of patients, usually in patients with severe heart disease. The worsened arrhythmia sometimes took the form of more intractable ventricular fibrillation. Heart block and bradycardia of an unacceptable degree also occurred in a small percentage of the patients. Hypothyroidism occurred in 2-5% of patients and hyperthyroidism in 0.5-2%. In addition to the relatively serious side effects, amiodarone causes a number of other effects, including nausea and vomiting (occurring in about 25% of patients) and neurological complaints (occurring in 20-40%) of patients.

Effectiveness

Because patients treated with amiodarone had life-threatening arrhythmias

that had been unresponsive to other therapy, placebo-controlled or active drug-controlled studies were not possible. Therefore, the clinical effectiveness data for amiodarone were derived from a sequential study that compared patients' responses to therapy with their pre-treatment baseline. Although this design is usually not optimal, several aspects of the study made it convincing.

Eighteen patients with ventricular arrhythmias were observed prior to the administration of amiodarone, on therapy, off therapy, and again on amiodarone. Of the 18, 13 had a recurrence of arrhythmia when amiodarone was stopped. The recurrent arrhythmias were, in 11 of the cases, ventricular fibrillation or ventricular tachycardia, generally the same as the arrhythmia for which amiodarone was originally given. Even in the two cases where ventricular tachycardia or fibrillation did not emerge, the ventricular ectopic beat frequency increased dramatically and may have been a "warning" arrhythmia. In 12 of the patients with a history of ventricular fibrillation or ventricular tachycardia, amiodarone was reintroduced and in 11 of those patients control was reestablished. Similar evidence of effectiveness was obtained from studies of patients in whom the dose was reduced. Of the patients with a history of syncope who remained on amiodarone, none was hospitalized for syncope and there was no episode of complete syncope.

Overall, the drug was considered fully effective in 60% of patients with life-threatening ventricular arrhythmias refractory to other antiarrhythmics, and partially effective (usually meaning that another drug had to be added) in most of the remaining patients.

Labeling

The package insert for amiodarone is unique in its detailed discussion of how to judge whether the drug is going to be effective and whether it should be continued. It contains a boxed warning about pulmonary toxicity and other serious risks and an extensive discussion

of interactions between amiodarone and other drugs, particularly warfarin, digoxin, quinidine and procainamide.

The labeling specifically states that the prescriber of amiodarone should have access to and familiarity with the full range of evaluatory procedures used in the care of patients with life-threatening arrhythmias and that the drug is indicated only after other antiarrhythmics have proven not to be useful, either because they were not effective or because they were not tolerated.

Estrogens For Osteoporosis

The use of oral short-acting estrogens in the lowest effective dosage to treat osteoporosis has been approved as an additional indication for certain drugs of this class, including conjugated estrogens, diethylstilbestrol, esterified estrogens, estradiol, ethinyl estradiol, estropipate and stilbestrol. Estrogen products used as contraceptives are not included in this approval.

The labeling of these products now states that short-acting estrogens are indicated in postmenopausal women with evidence of loss or deficiency of bone mass, to retard further bone loss and estrogen-deficiency-induced osteoporosis. Short-acting estrogens should be used with other measures that may retard osteoporosis, such as diet, calcium supplementation and exercise. A more favorable benefit/risk ratio for short-acting oral estrogens exists in a woman who has had a hysterectomy because she has no risk of endometrial carcinoma.

The labeling of short-term oral estrogens already states that while a postmenopausal woman not taking estrogens has a 1:1000 chance each year of contracting endometrial carcinoma, a woman taking estrogens has 5-10:1000 chances each year.

FDA recently recommended to estrogen manufacturers that they also note in their labeling that studies have shown the addition of a progestin for seven or more days of an estrogen cycle is associated with a lower incidence of

endometrial hyperplasia than an estrogen-only cycle. Morphological and biochemical studies of the endometrium suggest that 10 to 13 days of progestin are needed to provide maximal maturation of the endometrium and eliminate any hyperplastic changes. Whether this will provide protection from endometrial carcinoma has not been clearly established. Possible additional risks that may be associated with the inclusion of progestin in estrogen replacement regimens include effects on carbohydrate and lipid metabolism.

Minimizing Diagnostic Nuclear Medicine Exposure To Embryo, Fetus, Infant

Recommendations to minimize diagnostic nuclear medicine exposure to the embryo, fetus and nursing infant have been published in a pamphlet by FDA.

The recommendations are:

- The *patient* who is or thinks she might be pregnant, or who is nursing, should be encouraged to give this information to her attending physician when the examination history is taken.
- The *nursing woman* should be requested to suspend breast feeding for an appropriate period of time following a nuclear medicine examination. The nuclear medicine physician or medical physicist can advise the nursing woman as to the length of time that nursing should be suspended.
- The *attending physician* can use nuclear medicine consultation request forms in non-emergency situations to record the pregnancy and nursing status of a woman of childbearing age and should request the patient to provide this information. If there is a medical emergency, if the nuclear medicine examination will contribute vital information to the diagnosis, and if there are no alternative methods for obtaining this information that would result in lower radiation exposure, then the examination generally should be performed regardless of the patient's pregnancy status but with particular attention to technical modifications of the procedure that will minimize radiation exposure.

- The *nuclear medicine* technologist, in the absence of information about the patient's pregnancy or nursing status, should be requested to ask the patient if she is or may be pregnant or if she is nursing. If the patient replies that she is or may be pregnant or is nursing, the technologist should notify the physician in charge.

- The *nuclear medicine physician* should be aware of appropriate alternatives prior to conducting a nuclear medicine procedure on a pregnant or potentially pregnant woman or one who is nursing, and should be prepared to consult with the attending physician on possible alternatives. These alternatives include:

- a. Requesting the use of a radionuclide that delivers a lower radiation dose or one that is less likely to cross the placental barrier than the radionuclide generally used, if the diagnostic objectives can still be met.

- b. In the case of a known pregnancy, assessing the possibility of deferring the examination until the pregnancy is concluded.

- c. In the case of a possible but unconfirmed pregnancy, deferring the examination that is not immediately needed until the pregnancy is ruled out.

- d. Cancelling the non-emergency examination once aware that the patient is or may be pregnant.

- e. Requesting the nursing woman to suspend breast feeding for the period of time that radioactivity is present in milk.

- f. Ascertaining the advisability of using other clinical modalities to diagnose the patient's condition.

The pamphlet also presents the rationale for each of the above recommendations.

These recommendations are supported by the American College of Nuclear Physicians, the American College of Radiology, the American College of Obstetricians and Gynecologists, the Society of Nuclear Medicine, and the American Academy of Pediatrics.

Copies of the pamphlet can be obtained from the Office of Training and Assistance (HFZ-240), Center for De-

vices and Radiological Health, FDA, 5600 Fishers Lane, Rockville, MD 20857; phone (301) 443-2436.

The nuclear medicine pamphlet is part of the radiation recommendations series produced by FDA's Center for Devices and Radiological Health to provide guidance to medical professionals on ways to reduce radiation exposure. An earlier report from this series is entitled "Procedures to Minimize Diagnostic X-Ray Exposure of the Human Embryo and Fetus." Copies of this publication can be ordered from the National Technical Information Service, Springfield, VA 22161. The accession number is PB 82-110552. Price is \$9.95 paper, \$5.95 microfiche.

Request For Reports Of Disease After Immunization With Hib Polysaccharide Vaccine

As a part of the continuing evaluation of polysaccharide vaccine used for prevention of *Haemophilus influenzae* type b (Hib) disease, FDA requests that practitioners and health department personnel report all cases of Hib disease occurring after vaccination with Hib polysaccharide vaccine.¹ The short form on the back of this *Drug Bulletin* may be used for submitting reports to FDA, or the report may be made to the Centers for Disease Control or directly to the manufacturer of the vaccine. Currently there are three manufacturers licensed to produce the vaccine: Praxis Inc. (distributed by Mead Johnson Nutritional Division), Lederle and Connaught Inc.

This information is being requested as part of the routine continuing evaluation of vaccines. Hib vaccine is recommended for children who are at least 2 years old, although it may be considered for use in some children at 18 months of age who are considered at high risk. It is not recommended in children younger than 18 months old.^{2,3} (See also August 1985 *Drug Bulletin*.)

It is also important to report any se-